AQA

FUNCTIONAL SKILLS CERTIFICATE
Functional Mathematics

Level 2

Mark Scheme

4368
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Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

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Glossary for Mark Schemes

Examinations are marked to award positive achievement.

Marks are awarded for demonstrating the following interrelated **process skills**.

Representing Selecting the mathematics and information to model a situation.

- **R.1** Candidates recognise that a situation has aspects that can be represented using mathematics.
- **R.2** Candidates make an initial model of a situation using suitable forms of representation.
- **R.3** Candidates decide on the methods, operations and tools, including ICT, to use in a situation.
- **R.4** Candidates select the mathematical information to use.

Analysing Processing and using mathematics.

- **A.1** Candidates use appropriate mathematical procedures.
- **A.2** Candidates examine patterns and relationships.
- **A.3** Candidates change values and assumptions or adjust relationships to see the effects on answers in models.
- **A.4** Candidates find results and solutions.

Interpreting Interpreting and communicating the results of the analysis.

- **I.1** Candidates interpret results and solutions.
- **I.2** Candidates draw conclusions in light of situations.
- **I.3** Candidates consider the appropriateness and accuracy of results and conclusions.
- **I.4** Candidates choose appropriate language and forms of presentation to communicate results and solutions.

In particular, individual marks are mapped onto the following **skills standards**.

Representing Making sense of the situations and representing them.
A learner can:

Ra Understand routine and non-routine problems in familiar and unfamiliar contexts and situations.

Rb Identify the situation or problems and identify the mathematical methods needed to solve them.

Rc Choose from a range of mathematics to find solutions.

Analysing Processing and using the mathematics.
A learner can:

Aa Apply a range of mathematics to find solutions.

Ab Use appropriate checking procedures and evaluate their effectiveness at each stage.

Interpreting Interpreting and communicating the results of the analysis.
A learner can:

Ia Interpret and communicate solutions to multistage practical problems in familiar and unfamiliar contexts and situations.

Ib Draw conclusions and provide mathematical justifications.

To facilitate marking, the following categories are used:

M Method marks are awarded for a correct method which could lead to a correct answer.

A Accuracy marks are awarded when following on from a correct method. It is not necessary to always see the method. This can be implied.

B Marks awarded independent of method.

ft Follow through marks. Marks awarded following a mistake in an earlier step.

SC Special case. Marks awarded within the scheme for a common misinterpretation which has some mathematical worth.

oe Or equivalent. Accept answers that are equivalent.

eg, accept 0.5 as well as $\frac{1}{2}$

Q	Answer	Mark	Comments
1(a)	48	B1 *Aa*	
	Additional Guidance		

Q	Answer		Mark	Comments
1(b)	**Alternative method 1**			
	$\frac{2}{16}$ or $\frac{1}{8}$ or 0.125 or $1 - \frac{2}{16}$ or $\frac{14}{16}$ or $\frac{7}{8}$ or 0.875 or 24 ÷ 16 or 1.5 or 16 ÷ 24 or 0.6... or 0.7		M1 *Ra*	
	24 – their $\frac{2}{16}$ × 24 or (1 – their $\frac{2}{16}$) × 24 or $\frac{14}{16}$ × 24 or 24 – their 1.5 × 2 or 24 – $\frac{2}{\text{their } 0.6...}$ or 24 – 3		M1 *Rb*	Fully correct method
	21		A1 *Aa*	
	Alternative method 2			
	200 ÷ 16 × 2 or 25	550 ÷ 16 × 2 or 68.75	M1 *Ra*	200 can be 250 or 100 or 553 or 2 or 1
	200 ÷ 24 or 8.3... and (200 – their 25) ÷ their 8.3...	550 ÷ 24 or 22.9... and (550 – their 68.75) ÷ their 22.9...	M1 *Rb*	200 can be 250 or 100 or 553 or 2 or 1 Fully correct method
	21		A1 *Aa*	
	Additional Guidance			
	The sum of any combination of 200, 250, 100, 2 and 1 may be used for 200			

Q	Answer		Mark	Comments
1(c)	**Alternative method 1**			
	16 × 10 or 160	24 × 10 or 240	M1 *Ra*	Number of cookies
	their 160 ÷ 4 or 40	their 240 ÷ 8 or 30	M1 *Ra*	Number of bags of cookies
	their 40 × 1.35 or 54	their 30 × 1.75 or 52.5(0)	M1 *Rb*	Selling price of bags of cookies Must be a number of bags, not a number of cookies
	their 40 × 0.02 (+ 19.5(0)) or 0.8(0) or 20.3(0)	their 30 × 0.02 (+ 19.5(0)) or 0.6(0) or 20.1(0)	M1 *Rc*	Cost of bags or total costs for cookies
	their 54 – their 0.8(0) or 53.2(0) or their 54 – their 20.3(0) or 33.7(0)	their 52.5(0) – their 0.6(0) or 51.9(0) or their 52.5(0) – their 20.1(0) or 32.4(0)	M1 *Rc*	Selling price of bags – cost of bags or selling price of bags – total cost Number of bags cannot be zero or one Must be from a different number of bags for large and small cookies if both attempted
	their 53.2(0) – their 51.9(0) or their 33.7(0) – their 32.4(0)		M1 *Aa*	Difference in profits
	1.3(0) and Yes		A2 *Ib Ib*	A1 1.3(0) or 53.2(0) and 51.9(0) or 33.7(0) and 32.4(0) A1ft Correct decision for their value with 1st, 2nd, 3rd and 6th M1 gained

Q	Answer		Mark	Comments
1(c)	**Alternative method 2**			
	16 × 10 or 160	24 × 10 or 240	M1 *Ra*	Number of cookies
	their 160 ÷ 4 or 40	their 240 ÷ 8 or 30	M1 *Ra*	Number of bags of cookies
	their 40 × 1.35 or 54	their 30 × 1.75 or 52.5(0)	M1 *Rb*	Selling price of bags of cookies Must be a number of bags, not a number of cookies
	their 54 – their 52.5(0) or 1.5(0)		M1 *Rc*	Difference in selling prices
	their 40 × 0.02 (+ 19.5(0)) or 0.8(0) or 20.3(0)	their 30 × 0.02 (+ 19.5(0)) or 0.6(0) or 20.1(0)	M1 *Rc*	Cost of bags or total costs for cookies
	their 1.5(0) – their 0.8(0) + their 0.6(0) or their 1.5(0) – their 20.3(0) + their 20.1(0)		M1 *Aa*	Difference in selling prices – difference in cost of bags or difference in selling prices – difference in total costs Number of bags cannot be zero or one Must be from a different number of bags for large and small cookies
	1.3(0) and Yes		A2 *Ib Ib*	A1 1.3(0) or 53.2(0) and 51.9(0) or 33.7(0) and 32.4(0) A1ft Correct decision for their value with 1st, 2nd, 3rd and 4th M1 gained

Q	Answer		Mark	Comments
1(c)	**Alternative method 3**			
	16 ÷ 4 or 4	24 ÷ 8 or 3	M1 *Ra*	Number of bags of cookies (1 batch)
	their 4 × 1.35 or 5.4(0)	their 3 × 1.75 or 5.25	M1 *Rb*	Selling price of bags of cookies (1 batch) Must be a number of bags, not a number of cookies
	their 4 × 0.02 (+ 19.5(0) ÷ 10) or 0.08 or 2.03	their 3 × 0.02 (+ 19.5(0) ÷ 10) or 0.06 or 2.01	M1 *Rc*	Cost of bags or total costs for cookies (1 batch) Allow 19.5(0) for 19.5(0) ÷ 10
	their 5.4(0) – their 0.08 or 5.32 or their 5.4(0) – their 2.03 or 3.37	their 5.25 – their 0.06 or 5.19 or their 5.25 – their 2.01 or 3.24	M1 *Rc*	Selling price of bags – cost of bags or selling price of bags – total cost (1 batch) Number of bags cannot be zero or one Must be from a different number of bags for large and small cookies if both attempted
	their 5.32 – their 5.19 or their 3.37 – their 3.24 or 0.13		M1 *Aa*	Difference in profits (1 batch)
	10 × their 0.13		M1 *Ra*	Difference in profits (10 batches)
	1.3(0) and Yes		A2 *Ib Ib*	A1 1.3(0) or 53.2(0) and 51.9(0) or 33.7(0) and 32.4(0) A1ft Correct decision for their value with 1st, 2nd, 5th and 6th M1 gained

Q	Answer		Mark	Comments
1(c)	**Alternative method 4**			
	16 ÷ 4 or 4	240 ÷ 8 or 3	M1 *Ra*	Number of bags of cookies (1 batch)
	their 4 × 1.35 or 5.4(0)	their 3 × 1.75 or 5.25	M1 *Rb*	Selling price of bags of cookies (1 batch) Must be a number of bags, not a number of cookies
	their 5.4(0) – their 5.25 or 0.15		M1 *Aa*	Difference in selling prices
	their 4 × 0.02 (+ 19.5(0) ÷10) or 0.08 or 2.03	their 3 × 0.02 (+ 19.5(0) ÷10) or 0.06 or 2.01	M1 *Rc*	Cost of bags or total costs for cookies (1 batch) Allow 19.5(0) for 19.5(0) ÷ 10
	their 0.15 – their 0.08 + their 0.06 or their 0.15 – their 2.03 + their 2.01 or 0.13		M1 *Rc*	Difference in selling prices – difference in cost of bags (1 batch) or difference in selling prices – difference in total costs (1 batch) Number of bags cannot be zero or one Must be from a different number of bags for large and small cookies
	10 × their 0.13		M1 *Ra*	Difference in profits (10 batches)
	1.3(0) and Yes		A2 *Ib Ib*	A1 1.3(0) or 53.2(0) and 51.9(0) or 33.7(0) and 32.4(0) A1ft Correct decision for their value with 1st, 2nd, 3rd and 6th M1 gained
	Additional Guidance			
	Adding costs to selling price can score a maximum of M4 eg (alt 1) 35.3(0) – 33.6(0) = 1.7(0) M1 M1 M1 M1 M0 M0 A0			

Q	Answer	Mark	Comments
2(a)	120 + 4 × 140 or 120 + 560 or 680	M1 *Ra*	
	£680	A1 *Ia*	Must see £ symbol SC1 £660 or £1060
	Additional Guidance		
	Mark holistically with 2(a) check		
2(a) Check	Reverse calculation eg (680 – 120) ÷ 4 = 140 or alternative method eg 120 + 140 + 140 + 140 + 140 = 680	B1ft *Ab*	ft their calculation Must reverse to 120 or 140 or 4 or 5 or 0
	Additional Guidance		
	Mark holistically with 2(a)		

<table>
<tr><th>Q</th><th colspan="3">Answer</th><th>Mark</th><th colspan="2">Comments</th></tr>
<tr><td rowspan="9">2(b)</td><td colspan="3">4.87 and 65.7</td><td>B1
Aa</td><td colspan="2"></td></tr>
<tr><td>their 65.7 ×
0.2(0)
or 13.14</td><td colspan="2">0.8</td><td>M1
Ra</td><td colspan="2">their 65.7 can be 47.1 or 83.1 or 39.2</td></tr>
<tr><td>their 65.7 –
their 13.14
or 52.56</td><td colspan="2">0.8 × their 65.7
or 52.56</td><td>M1
Aa</td><td colspan="2">actual fuel efficiency (f)
their 13.14 cannot be 20 or 0.2
Award if method seen for 20% in first M1</td></tr>
<tr><td>62 × 5 × 46
or 14 260</td><td colspan="2">62 × their 4.87 ÷
their 52.56
or 5.7...</td><td>M1
Ra</td><td>miles (m)</td><td>miles × cost ÷ fuel
their 4.87 can be 4.96
their 52.56 can be 65.7</td></tr>
<tr><td>their 14 260 ×
their 4.87 ÷
their 52.56</td><td colspan="2">5 × 46 ×
their 5.7...</td><td>M1
Rc</td><td>miles × cost ÷ fuel
their 4.87 can be 4.96
their 52.56 can be 65.7</td><td></td></tr>
<tr><td colspan="3">[1320, 1321.30] and Yes</td><td>A2ft
Ib Ib</td><td colspan="2">A1ft [1320, 1321.30]
A1ft Correct conclusion for their value with 2nd, 3rd and 4th M1 gained
Only ft their 4.87 and their 65.7</td></tr>
<tr><td colspan="6">Additional Guidance</td></tr>
<tr><td colspan="6">Using (4.87 and) 47.1 1st M1 9.42 2nd M1 37.68 1843.(...) and Yes B0 M4 A2ft</td></tr>
<tr><td colspan="6">Using (4.87 and) 83.1 1st M1 16.62 2nd M1 66.48 1044.(...) or 1045 and No B0 M4 A2ft</td></tr>
<tr><td colspan="6">Using (4.87 and) 39.2 1st M1 7.84 2nd M1 31.36 2214.(...) and Yes B0 M4 A2ft</td></tr>
<tr><td colspan="6">Using 4.96 (and 65.7) 1345.(...) or 1346 and Yes B0 M4 A2ft</td></tr>
<tr><td colspan="6">Using 4.96 and 47.1 1st M1 9.42 2nd M1 37.68 1877.(...) and Yes B0 M4 A2ft</td></tr>
<tr><td colspan="6">Using 4.96 and 83.1 1st M1 16.62 2nd M1 66.48 1063.(...) or 1064 and No B0 M4 A2ft</td></tr>
<tr><td colspan="6">Using 4.96 and 39.2 1st M1 7.84 2nd M1 31.36 2255.(...) and Yes B0 M4 A2ft</td></tr>
</table>

Q	Answer	Mark	Comments
2(c)	**Alternative method 1**		
	89 and 98 and 77 and 86 and 95 and 110 and 88 and 75 or 1(h) 29(min) and 1(h) 38(min) and 1(h) 17(min) and 1(h) 26(min) and 1(h) 35(min) and 1(h) 50(min) and 1(h) 28(min) and 1(h) 15(min)	M1 *Ra*	At least 6 correct
	$\frac{\text{their } 5}{8}$ or 0.625 or $\frac{\text{their } 3}{8}$ or 0.375	M1 *Rc*	
	$\frac{\text{their } 5}{8} \times 120$ or $(1 - \frac{\text{their } 3}{8}) \times 120$	M1 *Aa*	
	75 and No or $\frac{75}{120}$ and No (and 89 and 98 and 77 and 86 and 95 and 110 and 88 and 75 and 90)	A2 *Ib Ib*	A1 75 or $\frac{75}{120}$ (and 89 and 98 and 77 and 86 and 95 and 110 and 88 and 75 and 90) A1ft Correct conclusion for their value with all M marks gained

Q	Answer	Mark	Comments
2(c)	**Alternative method 2**		
	89 and 98 and 77 and 86 and 95 and 110 and 88 and 75 or 1(h) 29(min) and 1(h) 38(min) and 1(h) 17(min) and 1(h) 26(min) and 1(h) 35(min) and 1(h) 50(min) and 1(h) 28(min) and 1(h) 15(min)	M1 *Ra*	At least 6 correct
	$\frac{\text{their } 5}{8}$ or 0.625	M1 *Rc*	
	their 5 ÷ 8 or 0.625 and 85 ÷ 120 or 0.7(08...) or 0.7(1)	M1 *Aa*	Converts $\frac{\text{their } 5}{8}$ and $\frac{85}{120}$ to a comparable form
	0.625 and 0.7(08...) or 0.7(1) and No (and 89 and 98 and 77 and 86 and 95 and 110 and 88 and 75 and 90)	A2 *Ib Ib*	A1 0.625 and 0.7(08...) or 0.7(1) (and 89 and 98 and 77 and 86 and 95 and 110 and 88 and 75 and 90) A1ft Correct conclusion for their value with all M marks gained
	Additional Guidance		

Q	Answer	Mark	Comments
2(c)	**Alternative method 3**		
	89 and 98 and 77 and 86 and 95 and 110 and 88 and 75 or 1(h) 29(min) and 1(h) 38(min) and 1(h) 17(min) and 1(h) 26(min) and 1(h) 35(min) and 1(h) 50(min) and 1(h) 28(min) and 1(h) 15(min)	M1 *Ra*	At least 6 correct
	$\frac{85}{120} \times 8$	M2 *Rc Aa*	
	[5.6, 5.7] and 5 and No (and 89 and 98 and 77 and 86 and 95 and 110 and 88 and 75 and 90)	A2 *Ib Ib*	A1 [5.6, 5.7] and 5 (and 89 and 98 and 77 and 86 and 95 and 110 and 88 and 75 and 90) A1ft Correct conclusion for their values with all M marks gained

Q	Answer	Mark	Comments
3(a)	**Alternative method 1**		
	128 × 25 or 3200 or 53.3...(h) or 40 × 30 or 1200 or 20(h)	M1 *Ra*	
	their 3200 + their 1200 or 4400 or 73.3...(h)	M1 *Rb*	Must be two times added
	2 pm – 8.30 am – 20 – 20 or 5(h) 30(min) – 20 – 20 or 290 or 4.8...(h)	M1 *Aa*	
	their 4400 ÷ their 290	M1 *Rc*	their 4400 could be 3200 or 1200
	[15.1, 15.2] or 15	A1 *Aa*	May be implied
	16	A1ft *Ib*	Rounds up their [15.1, 15.2] or the answer to their calculation Must score M4
	Alternative method 2		
	128 × 25 or 3200 or 53.3...(h) or 40 × 30 or 1200 or 20(h)	M1 *Ra*	
	2 pm – 8.30 am – 20 – 20 or 5(h) 30(min) – 20 – 20 or 290 or 4.8...(h)	M1 *Aa*	
	their 3200 ÷ their 290 or 11.(0...) or their 1200 ÷ their 290 or 4.(1...)	M1 *Rc*	
	their 11.(0...) + their 4.(1...)	M1 *Rb*	Must be two numbers of cleaners added
	[15.1, 15.2] or 15 or 11.(0...) and 4.(1...)	A1 *Aa*	May be implied
	16	A1ft *Ib*	Rounds up their [15.1, 15.2] or the answer to their calculation Must score M4

Q	Answer	Mark	Comments
3(a)	**Alternative method 3**		
	2 pm – 8.30 am – 20 – 20 or 5(h) 30(min) – 20 – 20 or 290 or 4.8...(h)	M1 *Aa*	
	their 290 ÷ 25 or 11.6 or their 290 ÷ 30 or 9.6... or 9.7	M1 *Ra*	Standard rooms per cleaner or deluxe rooms per cleaner
	128 ÷ their 11.6... or 11.(0...) or 40 ÷ their 9.6... or 4.(1...)	M1 *Rc*	
	their 11.(0...) + their 4.(1...)	M1 *Rb*	Must be two numbers of cleaners added
	[15.1, 15.2] or 15 or 11.(0...) and 4.(1...)	A1 *Aa*	May be implied
	16	A1ft *Ib*	Rounds up their [15.1, 15.2] or the answer to their calculation Must score M4
	Additional Guidance		
	Allow decimal times for M marks only		

Q	Answer	Mark	Comments
3(b)	4 × 18 or 72 or 3 × 8 or 24 or 2 × 11 or 22 or 1 × 9	M1 *Ra*	Not implied by 127
	their 72 + their 24 + their 22 + their 9 or 127	M1 *Rb*	Allow one error
	72 + 24 + 22 + 9 = 127 and 127 ÷ 50 = 2.54 or 4 × 18 + 3 × 8 + 2 × 11 + 1 × 9 = 127 and 127 ÷ 50 = 2.54 or $\frac{72+24+22+9}{50} = 2.54$ or $\frac{4\times18+3\times8+2\times11+1\times9}{50} = 2.54$ or 2.54 × 50 = 127 and 72 + 24 + 22 + 9 = 127 or 2.54 × 50 = 127 and 4 × 18 + 3 × 8 + 2 × 11 + 1 × 9 = 127 or 2.54 × 50 = 127 and 127 – 72 – 24 – 22 – 9 = 0 or 2.54 × 50 = 127 and 127 – 4 × 18 – 3 × 8 – 2 × 11 – 1 × 9 = 0	A1 *Aa*	Do not allow division by 5 followed by division by 10
	Additional Guidance		
	Totals seen next to table but other incorrect method used scores zero		
	Only 127 ÷ 50 = 2.54 scores M0 M1 A0		

Q	Answer	Mark	Comments
3(c)	0.75 × 168 or 126	M1 *Ra*	Number of rooms
	2.54 × 365 or 927.1 or 2.54 × their 126 or 320.04 or 365 × their 126 or 45 990	M1 *Rb*	their 126 could be 168
	their 927.1 × their 126 or their 320.04 × 365 or their 45 990 × 2.54 or 116 814(.6) or 116 815	M1 *Rb*	Number of cartons their 126 could be 168
	their 116 814(.6) ÷ 240 or 486.7... or 487	M1 *Rc*	Number of boxes their 116 814(.6) can be 927.1 or 320.04 or 45 990
	their 487 × 12.6(0)	M1 *Aa*	Cost of boxes Must be number of boxes × 12.6(0)
	6136.(20) and No or 6132.(...) or 6133 and No	A2 *Ib Ib*	A1 6136.(20) or 6132.(...) or 6133 A1ft Correct conclusion for their value with 2nd, 3rd, 4th and 5th M1 gained
	Additional Guidance		
	Use of 360 days 6048.76 and No M5 A0 A1ft 6048.76 M5 A0 A0ft		

Q	Answer	Mark	Comments
4(a)	9 ≤ number ≤ 12 and cm	B1 *Ia*	May be on the diagram or within a product Correct units must be seen
	Additional Guidance		
4(b)	Net of a cuboid with 5 or 6 faces	B1 *Ia*	
	Edge ≥ 15.2 cm labelled in correct position at least once	B1 *Aa*	
	Fully correct net of cuboid with length and width of all 6 rectangles correctly labelled at least once	B1ft *Ia*	Only ft their floor length in (a) Appropriate edges must be labelled ≥ 15.2 cm SC2 Fully correct apart from 12.7 cm ≤ edges < 15.2 cm instead of edges ≥ 15.2 cm
	Additional Guidance		
	Ignore flaps throughout		

Q	Answer	Mark	Comments
4(c)	**Alternative method 1**		
	6 × 4 or 24	M1 *Ra*	area of window
	0.16 × 112 or 17(.92) or 18 and 0.25 × 112 or 28	M1 *Aa*	Allow 112 to be 112 + 6 × 4 or 136 or 112 – 6 × 4 or 88
	24 and 17(.92) or 18 and 28 and Yes	A2 *Ib Ib*	A1 24 and 17(.92) or 18 and 28 A1ft Correct conclusion for their values with M2 scored
	Alternative method 2		
	6 × 4 or 24	M1 *Ra*	area of window
	$\frac{\text{their } 24}{112}$ (× 100) or 0.21(4...) or 21(.4...)	M1 *Aa*	Allow 112 to be 112 + 6 × 4 or 136 or 112 – 6 × 4 or 88
	21(.4...) and Yes or 0.21(4...) and 0.16 and 0.25 and Yes	A2 *Ib Ib*	A1 21(.4...) or 0.21(4...) and 0.16 and 0.25 A1ft Correct conclusion for their value(s) with M2 scored
	Additional Guidance		
	2nd M1 Use of 136 for 112 Alt 1 0.16 × 136 or 21.(76) or 21.8 or 22 and 0.25 × 136 or 34 Alt 2 $\frac{\text{their } 24}{136}$ (× 100) or 0.17(6...) or 17.(6...)% Use of 88 for 112 Alt 1 0.16 × 88 or 14.(08) or 14.1 and 0.25 × 88 or 22 Alt 2 $\frac{\text{their } 24}{88}$ (× 100) or 0.27(2...) or 0.273 or 27.(2...)% or 27.3%		

Q	Answer	Mark	Comments
4(d)	$\frac{5}{9}(46 - 32)$ or $\frac{5}{9} \times 14$ or $\frac{5}{9}(85 - 32)$ or $\frac{5}{9} \times 53$	M1 *Rc*	Allow 0.55... or 0.56 for $\frac{5}{9}$
	7.7… or 7.8 or 8 or 29.4… or 29	A1 *Aa*	
	8 and 29	A1ft *Ib*	ft their 7.7… and their 29.4… correctly rounded if M1 A0 and two values seen
	Additional Guidance		
	Both values must need rounding for A1ft		
4(e)	2000 ÷ 71 or 28.(1…) or 28.2 or 28 × 71 = 1988 or 29 × 71 = 2059	M1 *Aa*	
	28	A1 *Ib*	Embedded answer scores M1 A0
	Additional Guidance		
	Mark holistically with 4(e) check		
4(e) Check	Reverse calculation eg1 28.(1...) × 71 = 2000 eg2 2000 ÷ 28 = 71.(4...) or alternative method	B1ft *Ab*	ft their calculation
	Additional Guidance		
	Mark holistically with 4(e)		

Q	Answer	Mark	Comments
4(f)	2000 ÷ 48 or 41.(6...) or 41.7 or 42 or 41 × 48 = 1968 or 41 × 49 = 2009 or 42 × 48 = 2016	M1 *Aa*	
	their 41 – their 28	M1 *Ib*	their 28 from (e) their 41 and their 28 may be decimals
	13	A1ft *Aa*	Only ft their 28 from (e) Allow full marks for 13 from 42 – 29
	Additional Guidance		